# Exact Organization of Density Matrices and Entanglement Structure in the Kitaev Spin Liquid

Chen-Chih Wang, Yi-Ping Huang, Sungkit Yip, arXiv:2605.18002

Chen-Chih Wang

NTHU

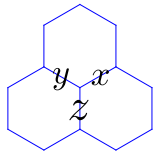
August 19, 2026

# Introduction and Motivation

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- Model Hamiltonian

$$H = -\sum_{\langle ij \rangle} J_{\alpha_{ij}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \quad ; \quad$$



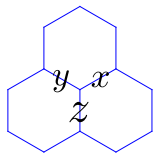

- Majorana fermion (parton) representation: $b^x$, $b^y$, $b^z$, $c \Rightarrow \sigma^\mu = ib^\mu c$
- Parton Hamiltonian: define $ib_i^{\alpha_{ij}} b_j^{\alpha_{ij}} = \hat{u}_{ij} \quad \Longrightarrow \quad H \to \sum_{\langle ij \rangle} iJ_{\alpha_{ij}} \hat{u}_{ij} c_i c_j$
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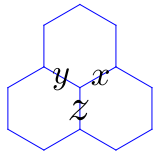
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- Illustrate the entanglement structure of the Kitaev model more clearly.
  - Directly solve the **entanglement spectrum (ES)**.
  - Do **NOT** use the **replica trick**.

# Results

- (Reduced) Density matrix directly reflects the fermionic feature.

$$\rho = \frac{1}{2^{N/2}} \hat{\Pi} \sum_{\mathrm{EP}(\mathfrak{D})} \mathfrak{C}_{\mathfrak{D}} \Sigma_{\mathfrak{D}}$$

$$\rho_A = \frac{1}{2^{(|\partial A|/2-1)}} \left( \frac{1}{2^{N^A/2}} \hat{\Pi}_A \sum_{\mathrm{EP}(\mathfrak{D}) \subset A} \mathfrak{C}_{\mathfrak{D}} \Sigma_{\mathfrak{D}} \right)$$

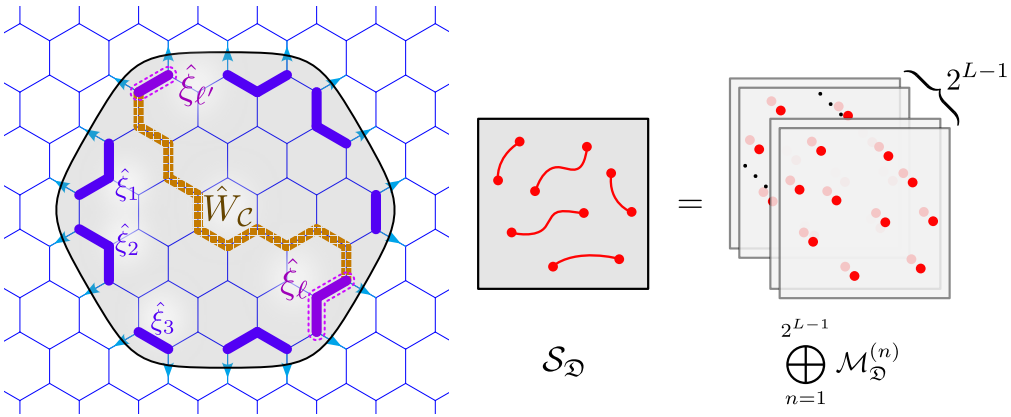
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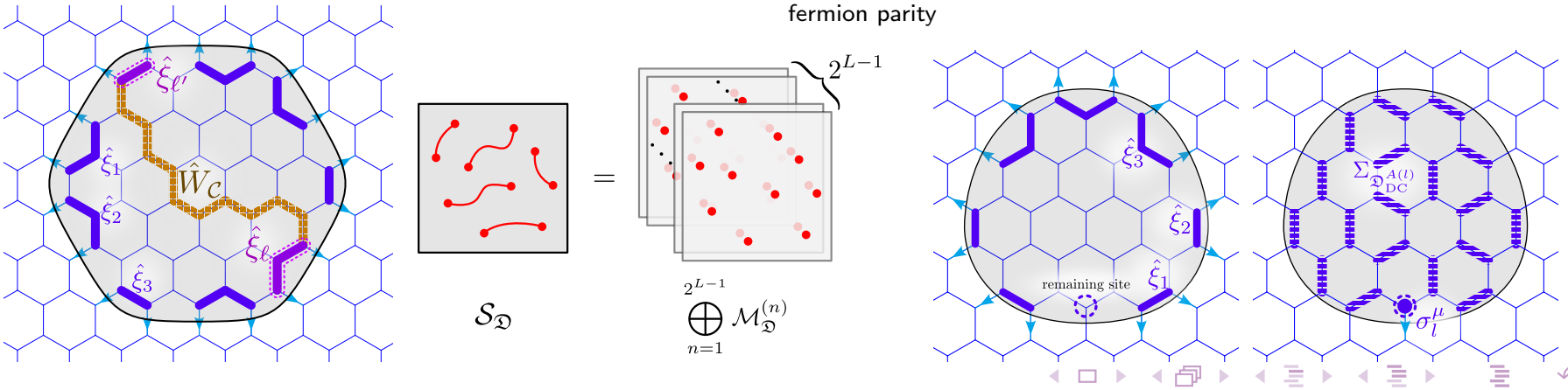
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- Subsystem fermion parity and entanglement in **odd** cases: Borrow a fermion from the gauge sector

$$\hat{\Xi}_A \equiv \sigma_l^\mu \Sigma_{\mathfrak{D}^{A(l)}_{\mathrm{DC}}} \to \underbrace{\left( b_l^\mu \prod_{i \in A} c_i \right)}_{\text{fermion parity}}, \qquad [\hat{\Xi}_A, \rho_A] = 0$$



# Concluding Remarks and Outlook

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- Expressing the density matrix in terms of the physical spin basis provides a natural starting point for **open-quantum-system**, **quantum-information**, and **quantum-resource** questions formulated in terms of **physical spin operations**.
- The stabilizer Rényi entropy calculation implies that the **magic** is governed only by the matter sector

$$M_\alpha\left(|\psi\rangle\right) := \frac{1}{1-\alpha}\ln\left(\sum_{\mathsf{S}\in\tilde{\mathcal{P}}_N}\frac{1}{2^N}\mathrm{Tr}\left(\rho\mathsf{S}\right)^{2\alpha}\right) = \frac{1}{1-\alpha}\ln\left(\frac{1}{2^{N-N_p}}\sum_{[\mathfrak{D}]\in\mathbf{G}}\mathfrak{C}_{\mathfrak{D}}^{2\alpha}\right)$$